

MLOps Assignment-5

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GitHub Branch: [Assignment-5](#)

Hugging Face Model: [ByteMaster01/ViT-S-LoRA-CIFAR100](#)

WandB Project: [MLOps-Assignment-5](#)

Overview

This report summarizes the implementation and results for Assignment-5:

- **Q1:** ViT-S / DeiT-S finetuning on CIFAR-100 with baseline head-only finetuning, LoRA sweeps, and the optional partial-unfreeze LoRA setup.
- **Q2:** Adversarial attacks and adversarial detection on CIFAR-10 using ResNet18, ResNet34, and IBM ART.

Final Result Snapshot

Experiment	Model	Metric	Result
Q1 Baseline	DeiT-S head-only	Test Accuracy	73.63%
Q1 Best LoRA	DeiT-S + LoRA ($r = 8, \alpha = 4$)	Test Accuracy	85.32%
Q1 Step-7 Optional	Partial-unfreeze LoRA ($r = 8, \alpha = 4$)	Test Accuracy	85.18%
Q2 Clean	ResNet18	Test Accuracy	93.23%
Q2 PGD Detector	ResNet34	Test Accuracy	100.00%
Q2 BIM Detector	ResNet34	Test Accuracy	99.99%

Q1: ViT-Small / DeiT-Small on CIFAR-100

Setup

For Q1, the pre-trained base model was `facebook/deit-small-patch16-224`. The classifier was adapted for CIFAR-100 with the following experiment groups:

- baseline head-only finetuning without LoRA
- LoRA applied to attention query, key, and value matrices
- ranks $\{2, 4, 8\}$
- alpha values $\{2, 4, 8\}$
- dropout 0.1
- optional step-7 partial-unfreeze experiment using the best LoRA configuration

Q1 Summary Table

LoRA layers	Rank	Alpha	Dropout	Test Acc (%)	Trainable Params
without	-	-	0.1	73.63	38,500
with	2	2	0.1	84.94	93,796
with	2	4	0.1	84.51	93,796
with	2	8	0.1	84.27	93,796
with	4	2	0.1	85.01	149,092
with	4	4	0.1	85.01	149,092
with	4	8	0.1	84.93	149,092
with	8	2	0.1	85.01	259,684
with	8	4	0.1	85.32	259,684
with	8	8	0.1	85.25	259,684
with (partial-unfreeze last 2)	8	4	0.1	85.18	222,820

Best observed configuration: $r = 8$, $\alpha = 4$ with **85.32%** test accuracy.

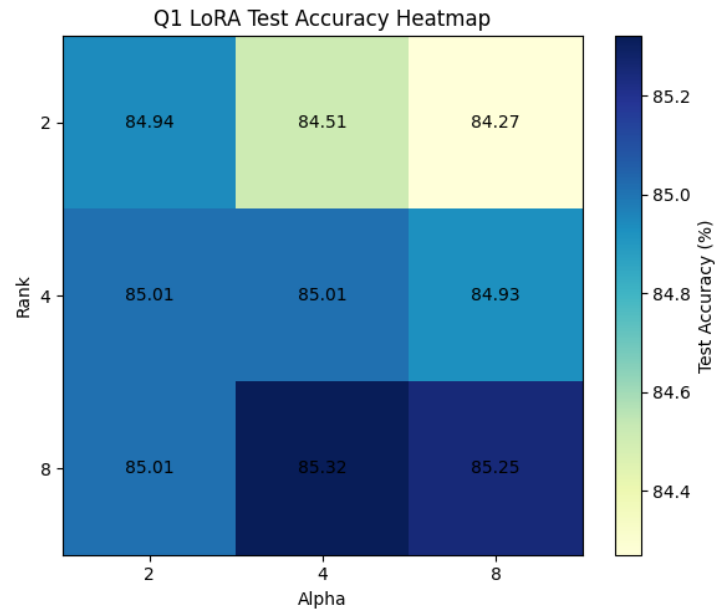


Figure 1: Q1 LoRA test accuracy heatmap over the rank/alpha sweep.

Baseline Head-Only Finetuning

Epoch	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)
1	1.5442	1.0790	64.17	70.05
2	0.9458	0.9854	73.36	71.73
3	0.8418	0.9583	75.76	72.39
4	0.7773	0.9402	77.46	72.79
5	0.7342	0.9340	78.66	73.10
6	0.7025	0.9240	79.58	73.36
7	0.6761	0.9201	80.35	73.38
8	0.6581	0.9174	80.87	73.61
9	0.6465	0.9136	81.17	73.63
10	0.6394	0.9129	81.46	73.60

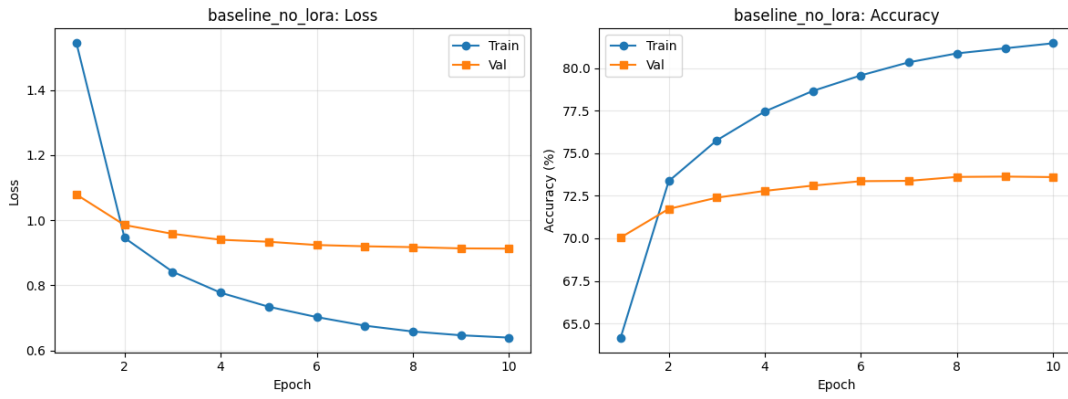


Figure 2: Baseline head-only train/validation curves.

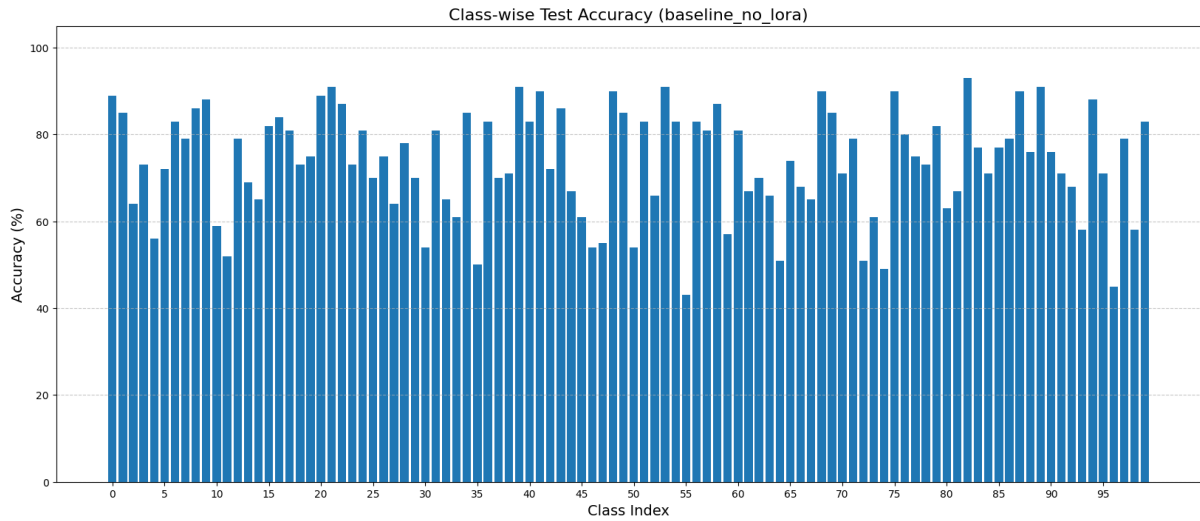


Figure 3: Baseline class-wise test accuracy histogram.

Best LoRA Configuration: $r = 8, \alpha = 4$

Epoch	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)
1	0.8802	0.5723	76.08	82.24
2	0.4407	0.5228	86.03	83.88
3	0.3294	0.5149	89.46	84.19
4	0.2427	0.5149	92.32	85.13
5	0.1744	0.5138	94.64	85.03
6	0.1242	0.5307	96.55	85.14
7	0.0899	0.5411	97.85	85.21
8	0.0672	0.5475	98.73	85.19
9	0.0546	0.5473	99.14	85.32
10	0.0482	0.5484	99.31	85.32

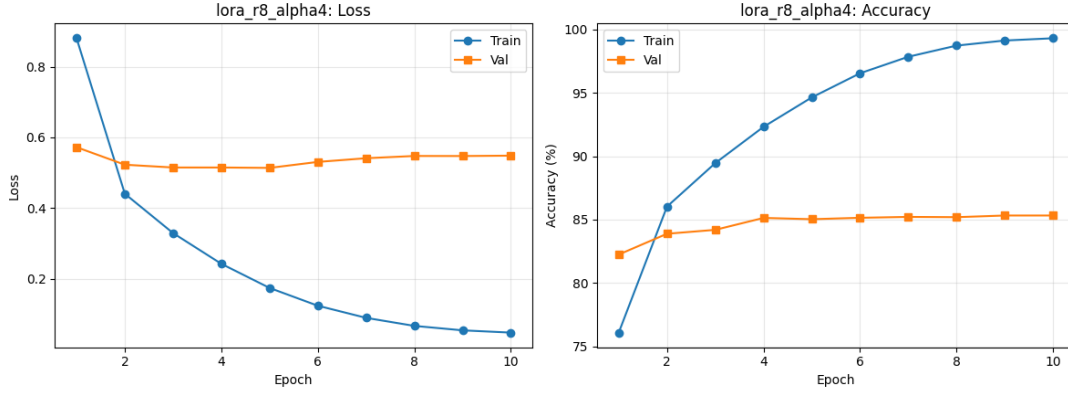


Figure 4: Best LoRA train/validation curves.

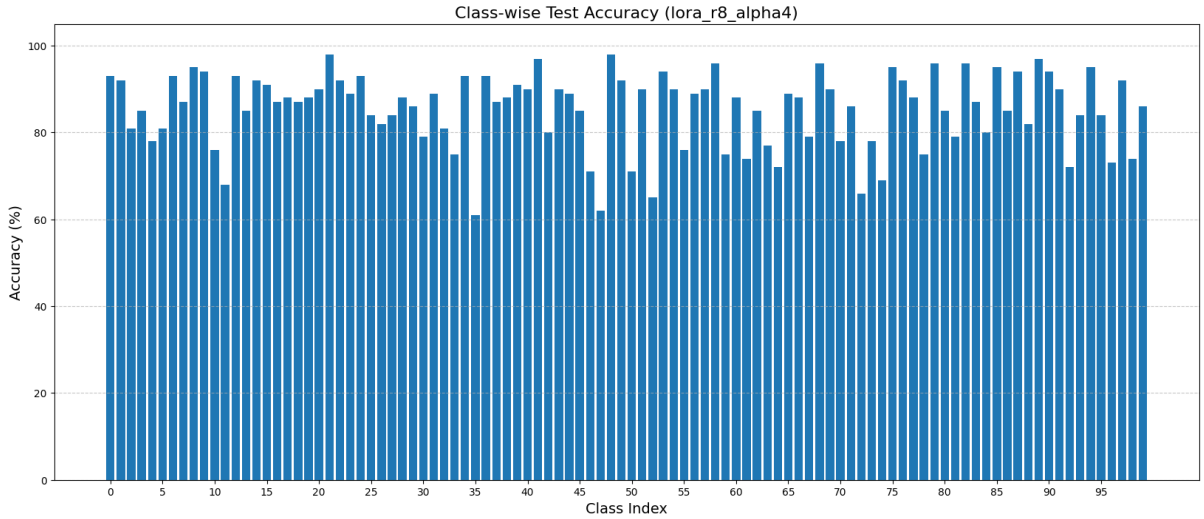


Figure 5: Class-wise test accuracy histogram for the best LoRA model.

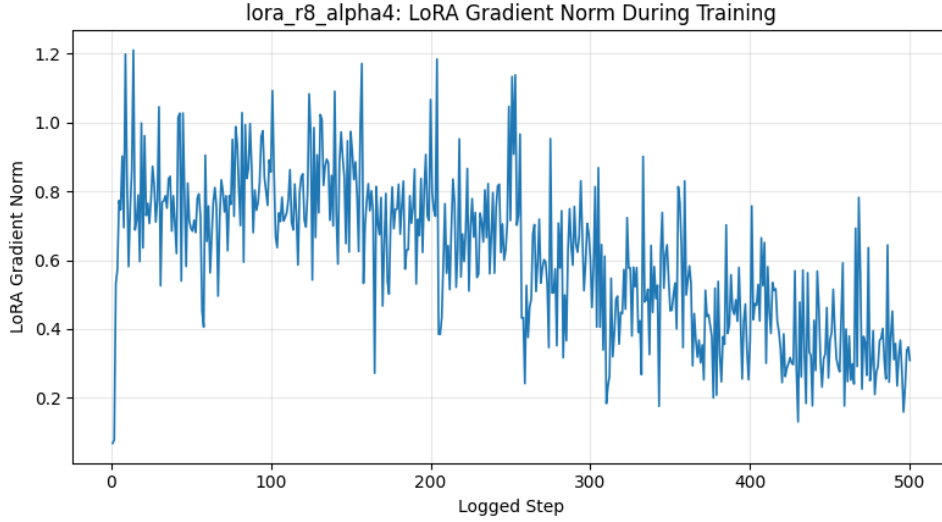


Figure 6: LoRA gradient norm during training for the best LoRA run.

Q1 analysis.

- The LoRA sweep produced a large gain over the baseline, increasing test accuracy from 73.63% to 85.32%.
- The best observed setup was rank 8, alpha 4.
- All LoRA runs outperformed the non-LoRA baseline.
- The gradient norm plot shows that the LoRA parameters received stable updates during training.

Optional Step-7 Partial-Unfreeze LoRA

Epoch	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)
1	0.9030	0.5660	75.76	82.53
2	0.4477	0.5278	85.91	83.94
3	0.3391	0.5137	89.29	84.33
4	0.2577	0.5158	91.90	84.81
5	0.1944	0.5250	94.05	84.71
6	0.1465	0.5274	95.79	84.93
7	0.1098	0.5338	97.33	85.00
8	0.0863	0.5389	98.18	85.10
9	0.0725	0.5414	98.61	85.04
10	0.0657	0.5409	98.92	85.18

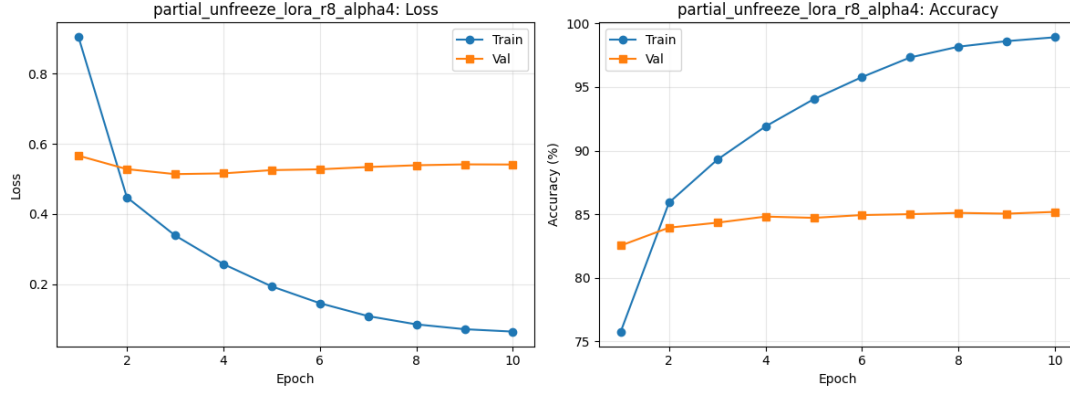


Figure 7: Partial-unfreeze LoRA train/validation curves.

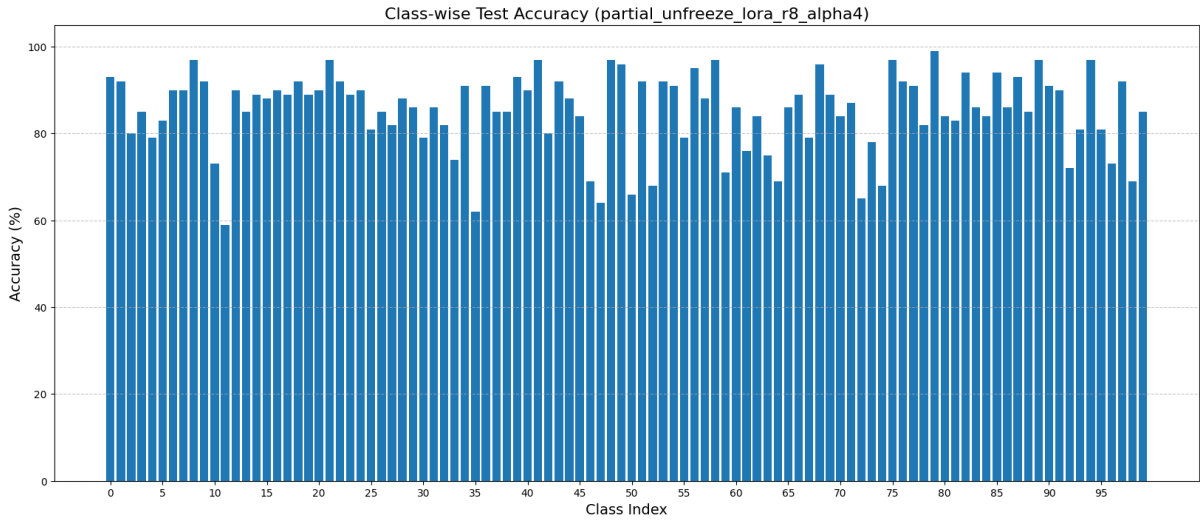


Figure 8: Class-wise test accuracy histogram for the partial-unfreeze LoRA run.

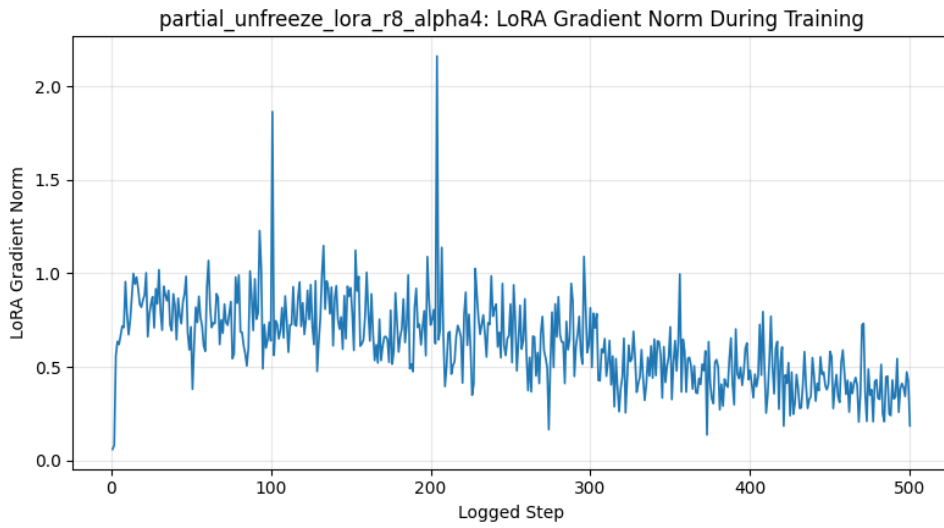


Figure 9: LoRA gradient norm during training for the partial-unfreeze LoRA run.

Step-7 observation.

The partial-unfreeze experiment achieved 85.18% test accuracy, which is very close to the best LoRA-only configuration but slightly lower than the best frozen-backbone LoRA result of 85.32%. In this run, partial unfreezing did not improve the final test accuracy.

Q2: Adversarial Attacks and Detection on CIFAR-10

Setup

Q2 contains:

- clean ResNet18 training on CIFAR-10
- FGSM attack from scratch
- FGSM attack using IBM ART
- PGD adversarial detector using ResNet34
- BIM adversarial detector using ResNet34

Clean ResNet18 Training

The clean ResNet18 model achieved **93.23%** test accuracy, which is well above the assignment requirement of 72%.

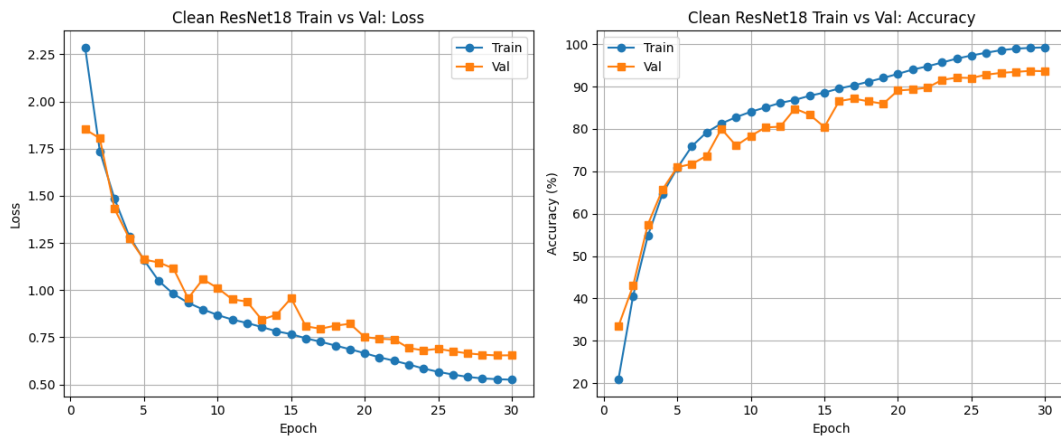


Figure 10: Clean ResNet18 train/validation curves on CIFAR-10.

FGSM: Scratch vs IBM ART

ϵ	Clean Acc (%)	Scratch Acc (%)	ART Acc (%)	Scratch Drop (%)	ART Drop (%)
0.00	93.23	93.23	93.23	0.00	0.00
0.05	93.23	43.00	46.01	50.23	47.22
0.10	93.23	14.25	14.92	78.98	78.31
0.15	93.23	10.93	11.30	82.30	81.93
0.20	93.23	10.61	10.83	82.62	82.40
0.25	93.23	10.43	10.31	82.80	82.92
0.30	93.23	10.33	10.35	82.90	82.88

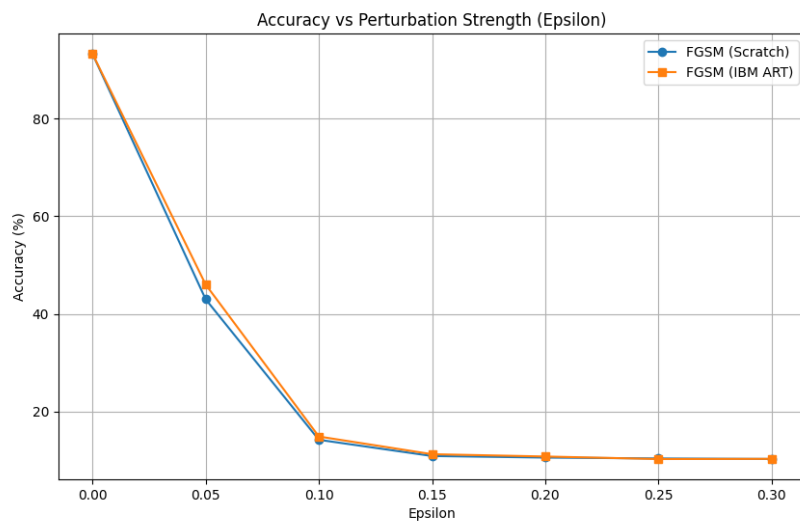


Figure 11: Accuracy vs perturbation strength for FGSM scratch and IBM ART FGSM.

FGSM Scratch (eps=0.1)



Figure 12: Clean vs adversarial samples for FGSM from scratch.

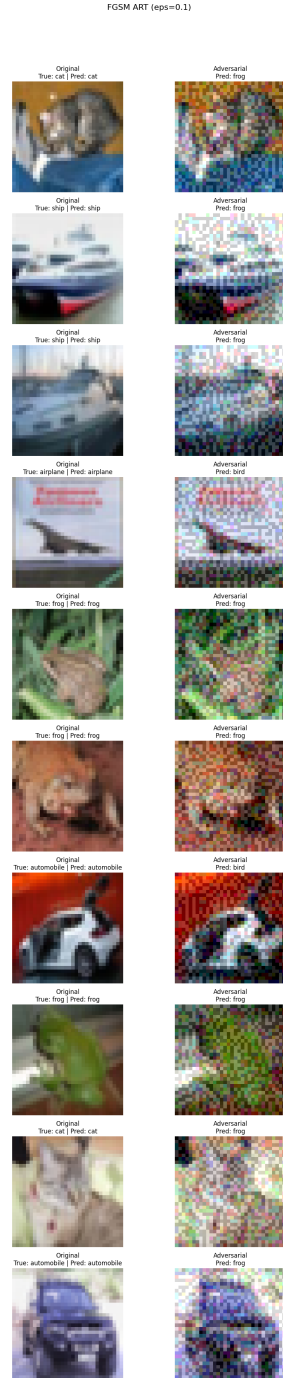


Figure 13: Clean vs adversarial samples for FGSM using IBM ART.

Q2 FGSM analysis.

- The clean classifier is strong enough that the attack impact is clearly visible.
- Both scratch FGSM and IBM ART FGSM show a steep reduction in classifier accuracy as epsilon increases.
- The scratch implementation closely matches the IBM ART implementation, indicating correct behavior.

Adversarial Detection with ResNet34

Attack	Best Val Acc (%)	Test Acc (%)	Meets 70% Target
PGD	99.99	100.00	True
BIM	100.00	99.99	True

PGD Detector

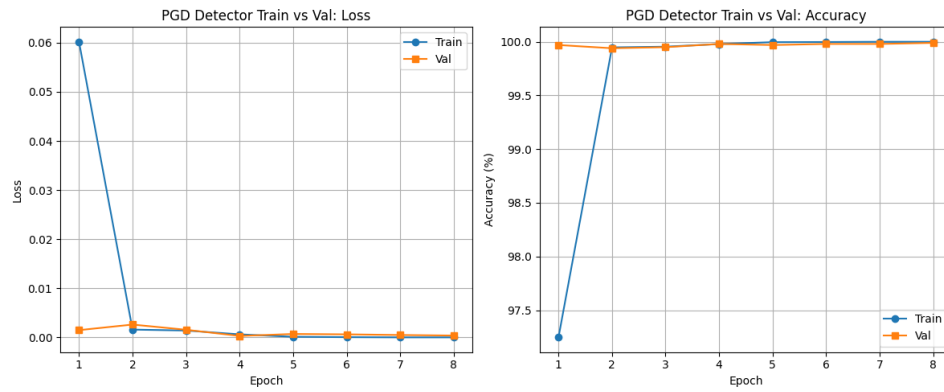


Figure 14: PGD detector train/validation curves.

PGD Adversarial Samples



Figure 15: PGD adversarial sample pairs logged for the detector experiment.

BIM Detector

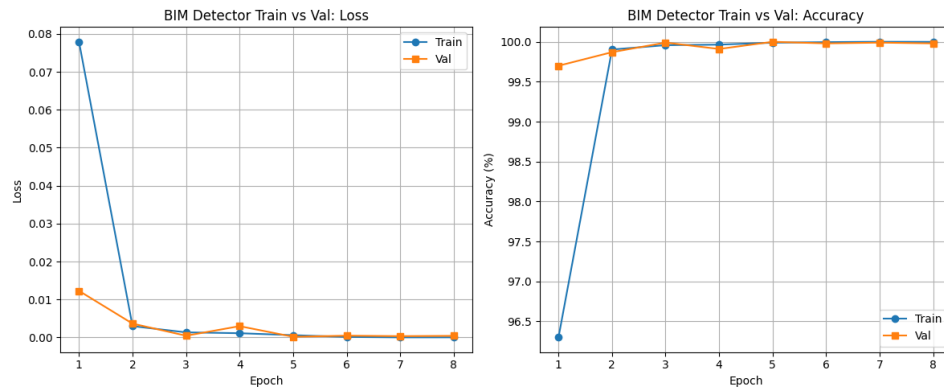


Figure 16: BIM detector train/validation curves.



Figure 17: BIM adversarial sample pairs logged for the detector experiment.

Q2 detector analysis.

- Both PGD and BIM detectors exceeded the assignment threshold by a large margin.
- PGD detection reached 100.00% test accuracy.
- BIM detection reached 99.99% test accuracy.
- The performance difference between PGD and BIM detectors is negligible in this run.

Artifact Locations

Q1:

- `output/q1_vit_lora/q1_test_results_table.csv`
- `output/q1_vit_lora/*_train_val_table.csv`
- `output/q1_vit_lora/*_train_val_curves.png`
- `output/q1_vit_lora/*_class_acc.png`
- `output/q1_vit_lora/*_lora_grad_norm.png`
- `output/q1_vit_lora/best_model/adapter_model.safetensors`

Q2:

- `output/q2_adversarial/clean_resnet18_train_val_table.csv`
- `output/q2_adversarial/fgsm_accuracy_comparison.csv`
- `output/q2_adversarial/detector_accuracy_comparison.csv`
- `output/q2_adversarial/*_detector_train_val_table.csv`
- `output/q2_adversarial/*.png`
- `output/q2_adversarial/resnet18_clean.pth`
- `output/q2_adversarial/detector_PGD.pth`
- `output/q2_adversarial/detector_BIM.pth`